

The Science-based Playbook of **Ecommerce Optimization**

Ariyh“

Boost your profits using the latest scientific discoveries in Ecommerce. No opinions, no sketchy data, no fluff.



Created by Thomas McKinlay,
Founder of Ariyh



With Stefano Puntoni,
Professor of Marketing at Wharton, University of Pennsylvania

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About this playbook

This playbook is a practical guide on how to apply the latest Ecommerce research from the world's best scientists. Use it to maximize your profits and get ahead of your competitors.

Most of these techniques have not yet been 'translated' from research into practice by books, courses, or consultants. This usually takes decades.

All science covered here has survived rigorous statistical analyses, been peer-reviewed (i.e. picked apart and challenged by other scientists), and been published in a reputable scientific journal.

This playbook **is not**:

- Full of filler information. It's straight to the point.
- Built on sketchy advice from marketing 'gurus'. This is science.
- A beginner's guide for Ecommerce. These are the latest optimization techniques.

Use this playbook to prioritize your optimization experiments, or as a guide to design a highly effective new store. Remember, always try to test methods in your own context before fully implementing them.

Want to stay in the loop with the latest science? Subscribe at no cost to Ariyh for a new recommendation every week. Just drop your email on [ariyh.com](mailto:brian@briancliette.me), if you haven't already.

Want personalized guidance to optimize your Ecommerce business? You can [reach out for an audit or a consultation](#).



What Ariyh readers say

“ Ariyh is a pilot's preflight checklist to elevate the success of your marketing. You would be mad to neglect it



Rory Sutherland - Vice Chairman
Ogilvy

“ An insanely valuable resource for any marketer worth their salt



Diane Wiredu - Founder
Lion Words

“ Ariyh does an excellent job at translating our academic research into clear and actionable advice.



Norbert Schwarz, Provost Professor
Marshall School of Business

“ Not only does it cover the latest, most relevant studies - it does so evenly, with both strengths and flaws of the research



Richard Shotton, Bestselling author
The Choice Factory

“ The missing link between academic papers and real-life practical applications



Sorin Patilinet - Senior Director Insights
Mars



Introduction

By Stefano Puntoni,
S. Kresge Professor of Marketing, The Wharton School, University of Pennsylvania

“ **Science is the best guide we have**

Too many marketers rely on biased opinions and distorted data. It's very easy to find someone that will give you advice. But what is it based on? If it's their personal experience, how do we know it was not simply chance? If they show data, how do we know it's accurate?

Scientific research is far from perfect. But it's orders of magnitude better than anything else we have. Use it right, and you will have a big edge over your competitors.

“ **Marketers struggle to keep up**

There's one common theme I see when I teach marketing managers and executives. No matter whether they work at giants like Google or Philips, or small family-run businesses: they struggle to keep up with what we've already discovered.

We spend billions of taxpayer money to fund advanced research, performed by some of the brightest minds in the world. But once it's published, instead of learning from it and using it, thousands of individual organizations waste resources trying to rediscover the same results. This is absurd. My hope is that this playbook, and the broader work of Ariyh, fix this.

“ **Don't forget your strategy**

This playbook gives you specific tactics that you can use to optimize your marketing efforts. But remember, great tactics without a strategy won't take you far. Make sure you have one in place.

I like to explain it using this metaphor: Imagine walking on very difficult terrain. To make sure that you don't fall, you need to look carefully at where you put your feet. These are your tactics. But if you are only looking down at your feet, you cannot look ahead and see whether you are going in the right direction. This is your strategy. If you don't look up, you are not going to fall today but sooner or later you will hit a wall.

Some of the institutions we should be grateful to for the research covered in this playbook:





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“ 1.

Website-wide Optimization Messaging

Focus on 3 main benefits



Recommendation

- Follow the rule of 3 when describing the benefits of your product or brand. Use three positive claims, not more, to maximize how persuasive your argument is.
- After three claims, people become skeptical, which undermines the message. In experiments, three positive claims were 10.4% more persuasive than four claims.



How it works

- Three is the smallest number where people start to identify patterns. Once a person feels they have identified a pattern, they are ready to use it to make a decision.
- When people know a message is trying to persuade (e.g. a brand's claim about their product), they are generally suspicious. The threshold of when that kicks in and starts to backfire starts after three positive claims.



Research

Scientific research by Shu, S.B. and Carlson, K.A. [When three charms but four alarms: Identifying the optimal number of claims in persuasion settings](#). *Journal of Marketing* (January 2014).

Universities: Anderson School of Management, University of California at Los Angeles, and McDonough School of Business, Georgetown University.

[Full Ariyh research summary](#)

PRO TIP: [Don't dilute your best features](#)

When describing your features, mixing good ones with your best ones [dilutes the impact of the best ones](#) - to the lowest common denominator. Stick to only talking about your best features.

PRO TIP: [Position rational messages up, emotional ones down](#)

In your visual layouts, [position rational messages](#) (e.g. about functionality) in the upper part of vertical spaces. Position emotional messages in the lower part. This plays to the metaphor people have of [the brain is up and the heart is down](#), and makes the message more effective.

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Human vs algorithm recommendations



Recommendation

- Product recommendations (e.g. You might also like...) are more or less effective depending on whether a human (e.g. Our expert's pick) or an algorithm gives it (e.g. Recommendation based on our data).
- If people are looking for the best practical value or functionality (e.g. a hotel stay for a businessperson), they prefer an algorithm-based recommendation.
- If they are buying for enjoyment or an experience (e.g. a hotel stay for a holiday), they prefer products recommended by humans.



How it works

- People have a prejudice that machines don't fully understand what people enjoy. But they trust algorithms and machines to better evaluate the pros and cons of functional items



Research

Scientific research by Longoni, C. and Cian, L. [Artificial intelligence in utilitarian vs. hedonic contexts](#). *Journal of Marketing* (November 2020).

Universities: Questrom School of Business, Boston University, and Darden School of Business, University of Virginia.

[Full Ariyh research summary](#)



Design

When to display products horizontally vs vertically



Recommendation

- Use a horizontal product layout (e.g. a carousel) if you want people to focus on quality and a strong primary feature (e.g. megapixels on a camera). People were up to 26% more likely to choose the highest quality - and more pricey - laptop (vs a vertical layout).
- Use a vertical layout (e.g. scroll down to see more products) if you want comparisons to focus on price or make people focus more on secondary features (e.g. a camera's screen size or weight).

How it works



- People approach decision-making in two main ways - the layout of a page can encourage people towards one or the other.
- Horizontal displays encourage abstract, high-level thinking leading people to consider a product's benefits, use, and quality.
- Vertical displays encourage concrete, practical, and detail-oriented thinking including how we can buy the item, leading to more price sensitivity.



Research

Scientific research by Mukherjee, S. [Vertical vs horizontal presentation formats result in different consumption choices due to construal priming](#). *Journal of Consumer Behavior* (May 2022).

Universities: Williams College of Business, Xavier University.

[Full Ariyh research summary](#)

PRO TIP: Structured vs unstructured design

Use [structured designs](#) (e.g. geometrical shapes, straight lines, symmetrical layouts) to make your brand feel more reliable and effective. Use unstructured design (e.g. unpredictable layouts, incomplete shapes, rounded lines) to make your brand seem more fun or exciting. In experiments, perfume ads received 39% more clicks when the design matched the brand positioning.



Trust and Loyalty

Show your costs to build trust



Recommendation

- Explain and display your costs for making your product. People will be more willing to buy. For example, sales were 22% higher when a \$115 wallet had an infographic alongside it showing its costs.
- Break down your company costs on a per-product basis. Be detailed by category (e.g. salaries, product development, shipping, packaging), but if categories start to become overwhelming (e.g. more than 8) group them together (shipping and packaging).
- Don't worry, people know you also need to make some profit. The effect holds for profit margins as high as 55%.



How it works

- People trust a company more if it discloses its product costs. This makes people like you more and more willing to buy from you.



Artisanal Chocolate
\$ 7.99
Add to Cart

Our Costs
Cocoa: \$1.29
Cane Sugar: \$0.10
Organic Butter: \$0.64
Packaging & Labor: \$3.26
Cost per Bar: \$5.29



Research

Scientific research by Mohan, B., Buell, R.W. and John, L.K. [Lifting the veil: The benefits of cost transparency](#). *Marketing Science* (April 2020).

Universities: University of San Francisco and Harvard Business School, Harvard University.

[Full Ariyh research summary](#)

Show off that you're a small company



Recommendation

- If you're a small business, in most cases you should explicitly communicate it and proudly show it off. People will perceive your products to be of higher quality - unless your products are high-tech (e.g. AI software, smartphones).
- If you're a small business making high-tech products, you can counter the negative effect by emphasizing the resources you spend on R&D.
- If you're a large business making low-tech products, showcase motivated employees through employee testimonials and pictures that showcase their passion for their work.



How it works

- Employee motivation is an important tool for how people judge companies. They assume employees at small companies are more motivated and passionate about their work, resulting in higher quality products.
- For high-tech products, investments in R&D are a key signal for product quality. People assume a larger company has more resources to invest, resulting in better products.



Research

Scientific research by Woolley, K., Kupor, D. and Liu, P.J. [Does company size shape product quality inferences? Larger companies make better high-tech products, but smaller companies make better low-tech products.](#) *Journal of Marketing Research* (August 2022).

Universities: SC Johnson School of Business, Cornell University; Questrom School of Business, Boston University, and Katz Graduate School of Business, University of Pittsburgh.

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PRO TIP: Offer extraordinary warranties

[Extraordinary warranties](#) (e.g. 5 years for a phone case) are an effective way to signal the quality and strength of your product. For example, people were 34.8% more likely to buy a hoodie with a 10-year warranty. The effect works best for people that don't know your brand and are familiar with the product category.

Auto-enroll customers in a simple loyalty program



Recommendation

- Give simple loyalty rewards to all your customers (e.g. Buy 10, get 1 free). In experiments, it increased customer value by 29.5% over 5 years.
- Register people automatically. The system does not need to be tiered. And remember, this affects lifetime value by increasing retention – the loyalty program may not show easily measurable results in the short or medium term.



How it works

- Customers who join a loyalty program feel more connected to a company when part of a reward program, and like the company for providing its rewards to its customers.
- Giving the reward after a threshold (e.g. after 10 sales) touches on people's desire for completeness. If someone has bought six items, they're more likely to buy the remaining four to complete the deal.



Research

Scientific research by Gopalakrishnan, A., Jiang, Z., Nevskaya, Y. and Thomadsen, R. [Can non-tiered customer loyalty programs be profitable?](#) *Marketing Science* (March 2021).

Universities: Jones Graduate School of Business, Rice University; Wharton School, University of Pennsylvania, and Olin School of Business, Washington University in St. Louis.

[Full Ariyh research summary](#)

PRO TIP: Make your rewards variable

When you give a reward - for loyalty or in a promotion - make it variable and uncertain (e.g. Buy 5 cups of coffee and get either 80% or 20% off your 11th purchase). Initially, fewer people may sign up for the program because of the uncertain reward, but those that do will repeat their actions often enough to quickly offset this drop.

Don't retarget customers too early



Recommendation

- Don't retarget customers with email reminders less than an hour after they abandon a product - it makes them less likely to complete a purchase compared to not receiving a reminder.
- Send an email reminder after 24 to 72 hours, to make customers more likely to return and complete their purchase.



How it works

- People become annoyed if they're retargeted too soon, especially if they were already planning to return.
- The memory of starting a purchase starts fading after a few days until eventually disappearing in a week. Reminders sent when people still remember why they wanted the purchase can increase the chances of them buying.



Research

Scientific research by Li, J., Luo, X., Lu, X. and Moriguchi, T. [The double-edged effects of e-commerce cart retargeting: does retargeting too early backfire?](#) *Journal of Marketing* (October 2020).

Universities: Nanjing University; Fox School of Business, Temple University; Fudan University, and Waseda University.

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Sustainability helps sales

- Customers [enjoy their experience more](#) when using a sustainable product - if they know it's sustainable. In experiments, people enjoyed writing with a sustainable pen 28% more and liked a song 12% more when listening on eco-friendly headphones.
- People are more likely to buy from a small business or startup with sustainable practices. [They trust them more and expect their products to be higher quality.](#) For example, people were 82% more likely to sign up for a newsletter promoting a business's products.
- A study of 10 million grocery sales, found that products that [highlight sustainability claims](#) (e.g. carbon neutral production, recyclable packaging, organic ingredients) have 6.4% higher revenue than those that don't.



Channel Optimization

Focus on the right channels, depending on your type of product



Recommendation

- Different channels have a more or less important role along your marketing funnel depending on the type of product you sell.
- If your products are for enjoyment (e.g. decorative candle) or an experience (e.g. travel), a strong social media presence and an appealing product page are relatively more important (other channels are still important, but relatively less).
- If your products are bought mainly for being functional and effective (e.g. insect-repellent candle, accounting software), a strong presence in search ads and third-party review and comparison websites is relatively more important.



How it works

- Pleasure-related products are a relatively impulsive process, where people seek a fun shopping experience. Curiosity, simple signals and appealing design have a strong effect. The journey often starts on social media before moving to product pages.
- Functional items are a relatively rational decision, where people seek the best value and the ability to easily compare features. This takes them to search engines and websites that help them compare products.

Product type	Crucial Channels	Customer Journey	Buying Decision
Bought for enjoyment or experience (e.g. chocolates, decor items, luxury goods, spas)	· Social media · Your product pages	· Impulsive and emotional · Seeking an enjoyable experience	Based on simple signals and appealing user experience
Bought for utility and function (e.g. detergent, appliances, insect repellent)	· Search engines · Comparison sites · Third-party reviews · Deal websites	· Rational and often pre-planned · Seeking easy comparisons to maximize value	Based on comparing products to find best value



Research

Scientific research by Li, J., Abbasi, A., Cheema, A. and Abraham, L.B. [Path to purpose? How online customer journeys differ for hedonic versus utilitarian purchases](#). *Journal of Marketing* (March 2020).

Universities: Universities: McIntire School of Commerce, University of Virginia and ComScore.

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Use marketplaces to build your initial customer base



Recommendation

- If you are relatively new or not well-known, marketplaces like Amazon or Taobao are a fast and cost-effective way to grow your sales.
- As customers discover you through the marketplace, some will start buying directly from you. On average, 1 of every 71 sales on a marketplace generates a new direct customer.
- Make the switch to buying from you as attractive as possible. Offer special promotions and additional product options on your own website. Use product packaging to help relay the message, but be careful to not get banned from the marketplace.



How it works

- Customer acquisition costs are significantly lower on large marketplaces (24% on average, including commissions), compared to acquiring direct customers (30% for mature brands, up to 100% for new brands starting out).
- Many customers will be ready to switch to your website if given a reason and the friction is low enough.



Research

Scientific research by Maier, E., and Wieringa, J. [Acquiring Customers through OnlineMarketplaces? The Effect of Marketplace Sales on Sales in a Retailer's own Channels.](#)
International Journal of Research in Marketing (September 2020).

Universities: HHL Leipzig Graduate School of Management and University of Groningen.

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Examples of Applications

Product	Context	Recommendation	Why
Affordable holiday package	Page promoting long weekend trip options	Show human recommendations of other similar packages (e.g. others also viewed)	For experiences, people prefer recommendations to come from humans, rather than algorithms
Travel charger	Website promoting a new super-fast portable charger	Design your page using straight lines, symmetry and geometric designs	Structured designs make the product on the page seem more reliable and effective
Handmade artisanal plates	Instagram page showcasing images of the most popular products	Include emotion arising sentences in images (e.g. "Eat with love") at the bottom of the image, not at the top	Messages related to emotions are more effective when shown at the bottom of a vertical space
Cybersecurity software for small businesses	Homepage explaining why the business is highly trustworthy	Show a breakdown of your main costs (e.g. how much you spend on research, external experts, employees)	People will appreciate your transparency, trust you more, and become more likely to buy.
Organic shampoo	Website selling shampoo and other beauty and toiletry items	Use rounded lines, incomplete shapes, and unpredictable patterns in your page design	Unstructured design makes a product seem more fun and exciting

“ 2.

Product Page Optimization

Social Proof

Show either number of sales or number of product views



Recommendation

- Show how many people have purchased or viewed your product. In experiments, purchase intentions increased up to ~58%.
- Start by showing views of your product, but as they increase, switch to showing purchase numbers. If your views-to-sales conversion rate is 5% or higher, show both. The thresholds to switch for views and purchases will vary depending on your product.
- Test different timeframes that reasonably suit your product. You can be quite flexible (e.g. today, this week, in the past 90 days).



How it works

- Product views and purchases are signals people use to judge the popularity and quality of a product. If a product has been viewed and bought often, it must be good.
- At high numbers, it's better to switch to sales numbers because high numbers of views (e.g. 15,600) become harder to mentally process, making them less effective.



Cashmere Scarf

\$ 79.99

Buy Now

Add to Cart

Viewed by **44** others today



Research

Scientific research by Das, G., Spence, M.T. and Agarwal, J. [Social selling cues: The dynamics of posting numbers viewed and bought on customers' purchase intentions](#). *International Journal of Research in Marketing* (January 2021).

Universities: Indian Institute of Management Bangalore; Bond University and Haskayne School of Business, University of Calgary.

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Keep showing up to 30% of sold-out product options



Recommendation

- When a particular product or product option (e.g. color) is sold out, mark it as sold out but keep displaying it on your page, instead of hiding it. This can increase sales up to 31%.
- This effect works on customers who haven't decided which specific product they want or those willing to buy an alternative.
- Make sure the proportion of sold-out options isn't more than 10%-30% of your products, or the effect may backfire. If it's more than 30%, hide some of the sold-out products.



How it works

- If a product is sold out, people see it as a signal of high quality – enough people wanted to buy it that it ran out.
- People assume that other similar products are also high quality, so they become more likely to buy them.



Research

Scientific research by Tian, J., Chen, R. and Xu, X. [A good way to boost sales? Effects of the proportion of sold-out options on purchase behavior](#). *International Journal of Research in Marketing* (April 2021).

Universities: School of Economics and Management, Tsinghua University; School of Management, Hainan University and Agricultural Bank of China.

[Full Ariyh research summary](#)



Product Page Design

Show multiple CTAs with different purchase quantities



Recommendation

- If your product can be naturally bought in multiple units (e.g. shampoo bottles, soda cans), instead of CTAs centered on whether to buy (Add to Cart), use multiple CTA buttons focused on how much to buy (e.g. Buy 1, Buy 2, Buy 3). In 37 experiments, this increased conversions 14% on average and sales up to 28%.
- Place multiple CTA buttons with different quantities next to your product. Start with three options (e.g. Buy 1, Buy 2, Buy 5), and test to refine how many buttons and what quantities to show.



How it works

- Because choices can be overwhelming, people usually divide buying into two decisions: “should I buy this?” and “how many should I buy?”.
- Focusing your messaging around the second pushes customers to think more about the later stages of decision-making, so they are more likely to buy and buy more.



Tortilla Chips

\$ 1.99

Buy 1

Buy 2

Buy 3



Research

Scientific research by Amir, O. and Duke, K.

[The importance of selling formats: When integrating purchase and quantity decisions increases sales.](#)
Marketing Science (May 2022).

Universities: Rotman School of Management, University of Toronto, and Randy School of Management, University of California at San Diego.

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Group similar product choices together, a few at a time



Recommendation

- When offering more than a handful of product options to customers (e.g. flavor, size, color, other attributes) present them one at a time in groups or 2-5 options based on similarities.
- For example, for a smartphone, first group technical options (screen size, how much memory) and then aesthetic options (color, matte/glossy finish) for customers to select.
- Think of how customers naturally make the decisions and try ordering your options accordingly. Generally, start with larger decisions (e.g. size, functionality) and move to smaller decisions (e.g. color, accessories).
- Make it easy for customers to go back and change their choices if they wish.



How it works

- When too many choices are shown at the same time, people feel a choice overload, making them leave or causing them to be unhappy with their choice.
- We use different types of criteria (e.g. price, aesthetics, practical use) when making different choices. When these different criteria have to be used at the same time, it increases choice overload.



Research

Scientific research by Dorn, M., Messner, C. and Wänke, M. [Partitioning the choice task makes Starbucks coffee taste better](#). *Journal of Marketing Behavior* (April 2016).

Universities: University of Bern and University of Mannheim.

[Full Ariyh research summary](#)

PRO TIP: Keep 'extra' options and add-ons simple and standardized

Make sure your pricing is [simple for people to understand](#) - especially when it comes to extra options, upgrades, or add-ons.

For example, for a cell phone plan, it's best to offer additional minutes as a simple to understand, rounded, price option (e.g. \$0.1/minute, not \$0.0831/minute). Using the same last digit between prices also helps (e.g. \$22 for Option A vs \$32 for Option A with an add-on)

[Flat rates plans](#) are another great way to do this. [68% of people in a study](#) were willing to pay more for a simpler but more expensive option.

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Product Descriptions

- Make sure your product language is [simple enough](#) that your target audience will understand it. Overly technical terms that your audience might not understand will make your product seem worse and significantly more expensive.
- When [describing colors](#), use emotionally evocative or funky names (e.g. mid-day sky blue) instead of traditional names (light blue). People will like your product more and be more likely to buy it.
- Use intuitive, [easy to understand measurements](#) (e.g. 20 washes of detergent, 8 soda cans). They are easier to understand than standardized measures (e.g. 200g of detergent, 96 oz. soda). Using measurements [people intuitively understand](#) (e.g. number of pieces instead of grams) can make people willing to pay up to 49% more for a product.
- Use [smaller units](#) to make your messages seem more precise and credible (e.g. describe a product's battery life as 180 minutes instead of 3 hours). This works particularly well for time, but can also work for other units (e.g. 32 oz. of coffee instead of 1 gallon).
- Highlight [branded ingredients or components](#) of your product (e.g. Nutella-containing cake, Gorilla smartphone glass). Demand increased up to 40% in experiments.
- If your product is bought for pleasure or fun (e.g. cookies), [don't use science-based messages](#) to describe it (e.g. lab-designed) - people will be up to 30% less likely to choose it. If your product is bought for its effectiveness (e.g. this playbook you are reading), science-based messaging can help (e.g. based on the latest science).
- If [how your product feels to the touch](#) is important, describe it (e.g. this jacket has a smooth outer layer with a soft and furry inner lining). Online clothes shoppers say not being able to feel materials is a key pain point. This helps overcome it.



Product Visuals

When using video is a must



Recommendation

- Use video, slideshows, or GIFs to display your product if people buy it for pleasure or excitement (e.g. fashion, holidays). In experiments, people were willing to pay 79% more.
- If people buy your product for its functionality (e.g. washing machine, drill), static images are all you need.



How it works

- Dynamic formats like videos, slideshows, and GIFs are more lifelike than static images and make it easier to imagine using a product.
- When people imagine using the product, they focus on the feelings the product gives and become less sensitive to higher prices.



Research

Scientific research by Roggeveen, A.L., Grewal, D., Townsend, C., & Krishnan, R. [The impact of dynamic presentation format on consumer preferences for hedonic products and services](#). *Journal of Marketing* (November 2015).

Universities: Babson College and University of Miami.

[Full Ariyh research summary](#)

PRO TIP: Use slow-motion shots

[Slow down action shots](#) of your product while it's in movement (e.g. a drink being poured). This technique works mainly for premium or luxury products bought for enjoyment. In experiments, it increased sales by 22%, ad clicks by 18%, and willingness to pay by 11%.

PRO TIP: Try shoppable videos

Social media-like shoppable videos are a growing opportunity to use video on your product pages. You can use them to show the product in action, explain its benefits more in detail, or even display video reviews. Try to make the video feel very personal and 'human', almost homemade (although still high quality).

One possible provider of shoppable video software is [Vidjet](#). I've had consistently good experiences with them, so I've negotiated a 20% lifetime discount for Ariyh readers. Just tell their support chat that you came from Ariyh, if you sign up for a paid plan.

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Where your models should be looking



Recommendation

- Models in your images should look away from the camera and the viewer if your product is bought for enjoyment (e.g. chocolate, a holiday). In an A/B test, sales of a sun hat were 30% higher and people rated the ad 19% better when the model was looking away.
- Models should look at the camera if the message is informational (e.g. product features, price) or involves negative emotions (e.g. poverty alleviation, anti-smoking).



How it works

- When a model's gaze looks away from the camera, it's easier for viewers to transport themselves into the model's place and imagine themselves using the product.
- A direct gaze makes it more clear that the person in the ad is someone else, not the viewer. It can also make the message feel more credible if the person giving the information is making eye contact.



Pleasurable product
Look away from viewer



Functional product
Looking at viewer



Research

Scientific research by To, R.N. and Patrick, V.M.
[How the eyes connect to the heart: The influence of eye gaze direction on advertising effectiveness.](#)
Journal of Consumer Research (February 2021).

Universities: Bauer College of Business,
University of Houston.

[Full Ariyh research summary](#)

PRO TIP: Show contextual background with women's clothes

If you are selling women's clothes, [show models wearing them while in a contextual background](#) (e.g. a reception for a cocktail dress, while jogging for sportswear), rather than a plain white background. They will be more likely to buy. This does not matter for men's clothes.

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Space out your products



Recommendation

- Increase the empty space between products in your page layout to make sure items are spaced out and not right next to one another.
- In experiments, this technique increased how much people liked and valued a product by 11.4% and almost doubled sales.



How it works

- When products are spaced out from one another, it's easier for people to delineate them and mentally process each object. This makes them seem more beautiful, so people value them more.



Research

Scientific research by Sevilla, J. and Townsend, C. [The space-to-product ratio effect: How interstitial space influences product aesthetic appeal, store perceptions and product preference.](#)
Journal of Marketing Research (October 2016).

Universities: Terry College of Business, University of Georgia, and School of Business Administration, University of Miami.

[Full Ariyh research summary](#)

PRO TIP: [Leave plenty of empty margins in ads](#)

In online ads, make sure to leave plenty of empty space at the margins. And if there are several products related to each other in the same ad, it's better to position them closer together in the center. In experiments, clicks [increased by up to 3x](#). Note: this was not tested on websites.

PRO TIP: [Show multiple copies of your product](#)

If people buy your product for one specific function, [show multiple copies](#) of it together (e.g. a product picture with 6 cans of energy drinks instead of a stand-alone can) to make the product seem up to 10.7% more effective at performing that one function (e.g. giving energy).

Make your products shine



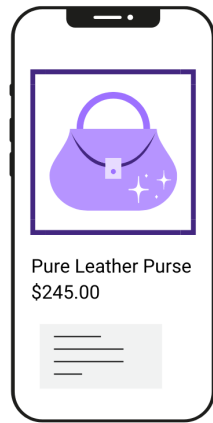
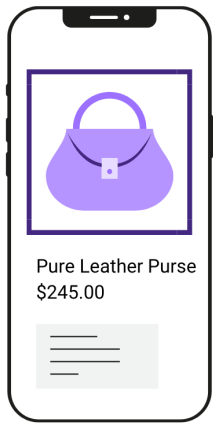
Recommendation

- Add a reflection on your product to make it look shiny. People found products 23% more attractive and said they were 32% more likely to buy them.
- When taking product photos, position your lighting so that it creates a reflection. Consider polishing the product. You can also digitally add in the reflection. This is especially important if your product's design is relatively simple and how it looks matters.



How it works

- People are attracted to shiny objects. It's a core human instinct developed by the need for clear, pure freshwater to survive.
- A shine also makes products look cleaner and newer, making them look better.



Research

Scientific research by Sharma, N. and Kumar, A.

[Drawn to the gloss: Examining the effect of product reflection on product aesthetics.](#)

Journal of Consumer Research (December 2022).

Universities: Dolan School of Business, Fairfield University, and Muma College of Business, University of South Florida.

[Full Ariyh research summary](#)

PRO TIP: Pretest your experiments using polls

To experiment faster and more effectively, pre-test your imagery and creatives using online polls. One possible platform I've had consistently good experiences with is [PickFu](#). If you want to try it, I negotiated 50% off your first poll. Just use the coupon ARIYH. Here are some [examples of polls](#) that you can run.

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Show a hand touching your product



Recommendation

- Show a hand touching or holding your product in a relevant way in product images (e.g. feeling a shirt, using a device, picking up a drink).
- The hand can be human or a digital lookalike, and should be shown from a first-person perspective for the viewer.
- In 15 experiments, people reported they were 16% more likely to buy a product. In a VR setting, the effect is even more powerful, with people willing to pay 32.5% more.



How it works

- People partially interpret the hand as their own. This increases their feeling of already owning the product, which makes them value it more.



Research

Scientific research by Luangrath, A.W., Peck, J., Hedgcock, W. and Xu, Y. [Observing product touch: The vicarious effect in digital marketing and virtual reality](#). *Journal of Marketing Research* (February 2022).

Universities: Tippie College of Business, University of Iowa; University of Wisconsin - Madison; University of Minnesota and University of California at Berkeley.

[Full Ariyh research summary](#)

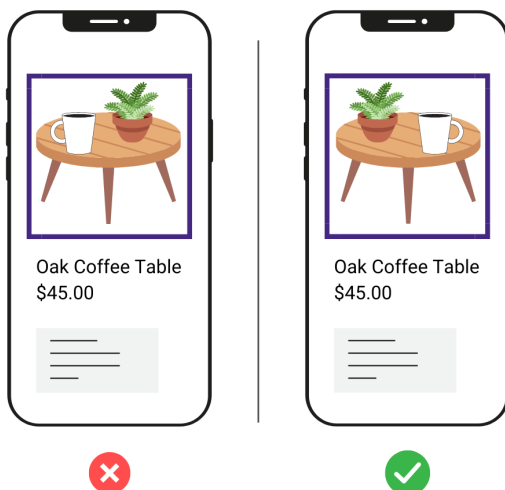
Include in images items that can be grabbed

Recommendation

- If your product can be held or grabbed (e.g. a mug, a remote, a screwdriver), position it around the right side of the image, so that people can more easily imagine picking it up and holding it with their right hand.
- If your product can't be picked up or held (e.g. a painting, software), include in the image another object that can be picked up, and place it near the main product.
- Signups to a supermarket loyalty program were 3x higher when a vegetable peeler was strategically positioned on the right side of the billboard.

How it works

- When people see an object positioned so it can be grasped by their dominant hand, they have a stronger feeling that they own it. The effect spills over to other items in the same field of vision.
- 90% of the world's population is right-handed, so you should tailor for them to maximize your sales.



Research

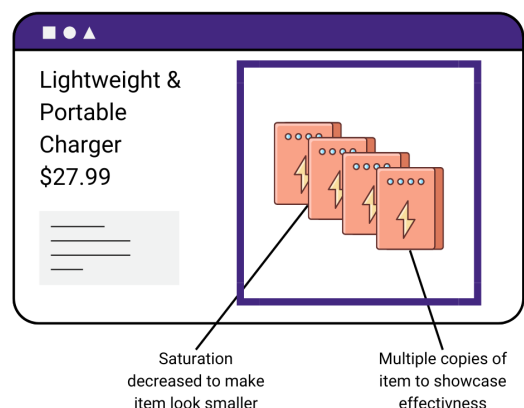
Scientific research by Maille, M., Morrin, M., Reynolds-McInay, R. [On the other hand...: Enhancing promotional effectiveness with haptic cues.](#) *Journal of Marketing Research* (October 2019).

Universities: KEDGE Business School; Rutgers University - Camden and Oregon State University.

[Full Ariyh research summary](#)

PRO TIP: Adjust your product's color saturation

The [color saturation of your product](#) makes it look bigger or smaller. If you want your product to look larger (e.g. a large bag of chips), use high color saturation. If you want your product to seem smaller (e.g. a portable charger), use low color saturation. People will be more likely to buy.



Surround your product with complementary products

Recommendation

- Surround your main product with complimentary products (e.g. accessories, add-ons). People will pay more attention to your main product and will be more likely to buy it. Sales of the complementary products aren't affected.
- The more complex and diverse the display of complementary products - including a mix of colors or shapes - the stronger the effect (e.g. a variety of sunglasses, necklaces, scarves, and sun hats alongside a dress for sale).

How it works

- Displays with high design complexity (different shapes, sizes, colors, and arrangements) tend to be more likable, interesting, and engaging.
- Using complimentary items (vs unrelated items) ensures that people aren't confused by the display, and are able to focus their increased attention on the main product.

Research

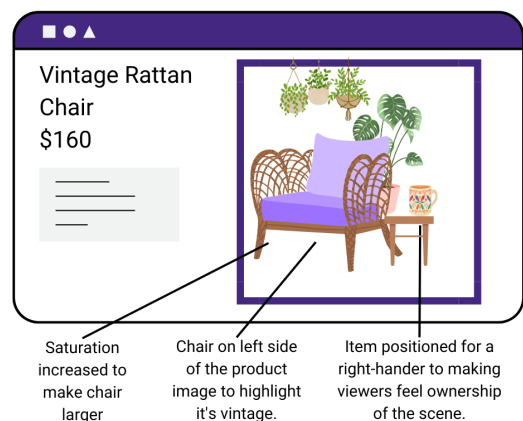
Scientific research by Radon, A., Brannon, D.C., and Reardon, J. [Ketchup with your fries? Utilizing complementary product displays to transfer attention to a focal product.](#) *Journal of Retailing and Consumer Services* (October 2020).

Universities: Södertörn University and Monfort College of Business, University of Northern Colorado.

[Full Ariyh research summary](#)

PRO TIP: Adjust the position of future or past-related products

People like future-oriented products (e.g. high-tech) more when they are in the right side of an image. Vice versa, they like past-related products (e.g. vintage) more when they're on the left. Flip this if your audience reads from right to left (e.g. Arabic, Urdu, Hebrew).





Examples of Applications

Product	Context	Recommendation	Why
Computer monitor for work	Marketplace selling mid-quality office hardware.	Use light reflection to add a shine to the monitor in product images	A shine will make the monitor seem newer and make people value it more
Hand-crafted leather sandals	Small business selling premium, handmade shoes	Showcase one of the shoemakers, and emphasize the small nature of the company	Products will feel higher quality, because small business workers seem more dedicated
Pure silk scarf	Product page showing and describing a high-end silk scarf	Describe how silky and smooth the scarf feels to the touch	People will feel they have a better understanding of the product, and one less objection to buy
High-end holiday in South Africa	Travel agency selling pre-planned premium packages	Models in your images and videos should look away from the viewer	Viewers will more easily imagine themselves in place of the model, so they'll be more likely to buy
Packet of Italian-brand pasta	Product page of an online groceries store	Show quantity-based CTA buttons (e.g. Buy 2, Buy 5)	Buyers will order more items, increasing the size of their order
2 hour Thai cooking class	Online booking page of a Thai chef offering private lessons	Keep showing options that are sold out. If sold-out options become more than 30%, start hiding those dates	The controlled scarcity increases quality perceptions of the class, making people feel constrained in their choices

“ 3.

Shipping and Returns Shipping Experience

Add a handwritten thank you note in your deliveries



Recommendation

- Give customers a hand-written thank you note with their purchase thanking them for their purchase. Include it in the packaging of their order.
- Both an original or a photocopy work equally well. It doesn't have to be personalized, although that might be a plus. The effect is especially strong on already loyal customers.
- The note should feel like a genuine thank you – don't combine it with any other promotions or discounts that may cloud the message.



How it works

- Handwritten notes add a human touch to the digital, contactless process of ordering online. These notes make you feel warm, caring, and sincere, making customers feel closer and more loyal to you.
- If there's an additional discount or promotion shared alongside the note, people call into question the sincerity of the note.



Research

Scientific research by Kim, S., Choi, J., and Kim, S.H. [Do handwritten notes benefit online retailers? A field experiment.](#) *Journal of Interactive Marketing* (July 2022).

Universities: Robert H. Smith School of Business, University of Maryland and School of Business, Yonsei University.

[Full Ariyh research summary](#)

PRO TIP: Scent your product

Add a [scent to your product](#) to make it more memorable to customers. You can scent the packaging or add it to the product itself. People remember your product for longer, so they might be more likely to recommend it to others, or buy it again if they find it in a store.

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Offer a free shipping subscription



Recommendation

- A subscription to access free shipping (e.g. Amazon Prime; pay a monthly fee for free deliveries) increases how much people spend with you and the variety of products they buy. In another study, [subscription members spent over 3x on a company's products](#).
- Considering requiring a minimum order threshold (e.g. \$20) to avail of free shipping. Otherwise, customers start to split their order into many smaller orders.



How it works

- People see a membership fee as a sunk cost, so they try to use the benefit of the membership as much as possible to maximize their investment.
- Without a minimum threshold for free shipping, there's no incentive for people to place large orders and instead place more orders, but of smaller quantities.



Research

Scientific research by Guo, F. and Liu, Y. [The effectiveness of membership-based free shipping: An empirical investigation on consumers' purchase behaviors and revenue contribution](#). *Journal of Marketing* (February 2023).

Universities: Poole College of Management, North Carolina State University, and Mays Business School, Texas A&M University.

[Full Ariyh research summary](#)



Returns

Free returns are a standard that people expect



Recommendation

- Free, no-questions-asked returns are a standard that customers expect from you. While costly in the short-term, in the long-term people spend more per order and buy a greater variety of products.
- Make your return process as simple and easy as possible. Include a return label, or provide an option to print one from your site. Ideally, allow customers to return the item in the same packaging.
- Offer options for customers to drop off returns either at your physical stores or in partnership with a logistics company at conveniently located drop-off points.



How it works

- When people know they can easily return a product, they're more willing to buy it even if they are uncertain about whether they will keep it or whether it will fit.
- A generous return policy is also seen as a signal of quality on the part of a brand – if a brand was not confident in the quality of their product, they wouldn't allow customers to easily return it.



Research

Scientific research by Patel, P., Baldauf, C., Karlsson, S., and Oghazi, P. [The impact of free returns on online purchase behavior: Evidence from an intervention at an online retailer.](#)
Journal of Operations Management (March 2021).

Universities: Villanova School of Business, Villanova University; Stockholm Business School, Stockholm University; Lund University and Södertörn University.

[Full Ariyh research summary](#)

PRO TIP: Customer benchmark you against Amazon

Whether you like it or not, people [judge online shopping experiences](#) against Amazon - and usually others fall short. Most people highlight product returns, exchanges, and refunds as the main areas they are unhappy with non-Amazon sites, so focus on making these processes as easy, simple, and fast as possible.

Offer instant refunds on returns



Recommendation

- Give an instant refund as soon as customers ship back a product. This improves satisfaction and loyalty, without increasing how likely people are to return a product.
- To avoid abuse, you can limit instant refunds to products that are unlikely to need a manual examination after a return (e.g. a book)



How it works

- Unlike returns in physical stores, online returns usually have a waiting period between returning the item and receiving the refund. This waiting triggers anxiety in people, as they don't know when the refund will be given.
- People also tend to judge how fair a process is by how fast and straightforward it is. Since an instant refund is instantaneous and simple, people view it as a fairer process and a signal that the company cares about its customers.

Item for Return



Return Options

- ☐ Instant refund on your credit card
- ☐ Replace the item
- ☐ Exchange for a different size or color



Research

Scientific research by Martinez-Lopez, F.J., Feng, C., Li, Y. and Lopez, D. [Using instant refunds to improve online return experiences](#). *Journal of Retailing and Consumer Services* (July 2022).

Universities: EAE Business School; University of Granada; Zhejiang University of Technology, and Esade Business School.

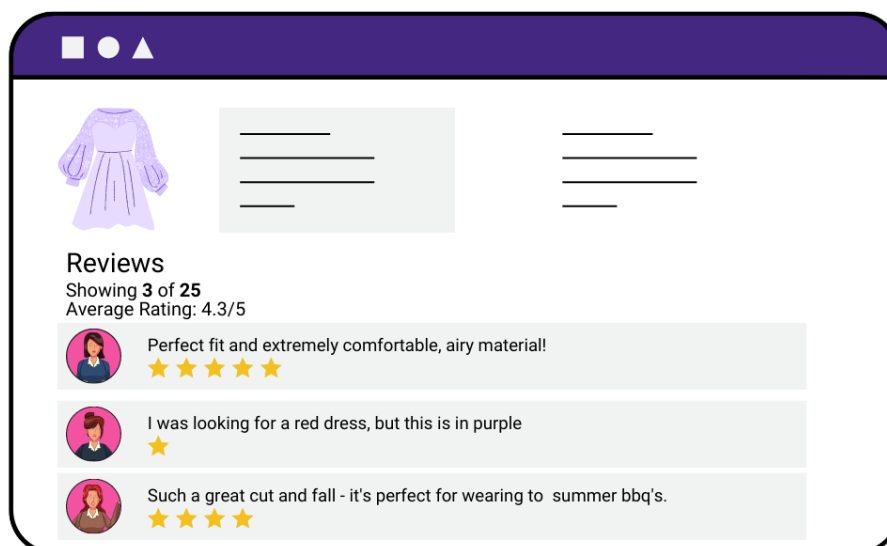
[Full Ariyh research summary](#)

“ 4.

Reviews and Complaints

Which Reviews to Show

- When you launch a new product, work hard to [get a first review that's positive](#) (e.g. ask early happy customers to leave a review as soon as possible). The first review biases the subsequent reviews. On Amazon and Best Buy, if the first review of a product was negative, the average star rating one year later was 0.29 lower than it should have.
- [Imperfect ratings are better than perfect ones](#) - which seem suspicious. Products with an average rating between 4 and 4.5 stars sell more than those with a rating between 4.5 and 5 stars.
- Some [negative reviews can be good for you](#), if they're irrelevant to most customers (e.g. that the product doesn't ship to a very specific area). For example, people rated an espresso machine 16% better when with four 5-star reviews, it also had an irrelevant 1-star review (that using different types of water impacted the coffee taste).





What and When to Ask

Ask for feedback using a positive framing



Recommendation

- Feedback surveys asking what customers enjoyed about your product increases future spending. People who responded to feedback spent 131% more than those who didn't.
- The effect is strongest when you start the survey with an open-ended question asking people to remember what they liked the most. But simply receiving the request for feedback already has a positive effect.
- Set up an automated system (e.g. via email) that automatically asks for feedback. Frame the first question positively, but you can follow that with more typical questions to improve your product's shortcomings.



How it works

- People tend to develop a better impression of the other party when they're asked for their opinions.
- Memory is easily influenced, so asking people to remember and explain positive experiences in open-ended questions leads them to remember the best parts of their experience. This makes positive memories of your product stronger.



Research

Scientific research by Bone, S. A., Lemon, K. N., Voorhees, C. M., Liljenquist, K.A., Fombelle, P.W., Detienne, K.B. & Money, R. B. [“Mere Measurement Plus”: How Solicitation of Open-Ended Positive Feedback Influences Customer Purchase Behavior](#). *Journal of Marketing Research* (February 2017).

Universities: Huntsman School of Business, Utah State University; Carroll School of Management, Boston College; Eli Broad School of Business, Michigan State University; Marriott School of Management, Brigham Young University and D'Amore-Kim School of Business, Northeastern University.

[Full Ariyh research summary](#)

When to ask for reviews



Recommendation

- Reminders to leave a review backfire if sent too early. In experiments, reminders to leave a review after a holiday reduced the number of reviews given if sent after only 24 hours of returning. The effect is particularly negative for younger people.
- The ideal timing to ask for a review for most products seems to be between 9 and 13 days after the purchase or experience. It might be sooner for products that can easily be immediately judged (e.g. paper towels, soda), and longer for products that take more time to evaluate (e.g. beauty treatment).



How it works

- Reminding someone to do a task jogs their memory of the experience. But if the reminder is sent when the experience is still fresh, the negative reaction outweighs the benefit.
- People value their independence and autonomy, and push back if they feel their freedom is being threatened. Being pushed to leave a review can make people feel their independence is being threatened, leading to a backlash.



Research

Scientific research by Jung, M., Ryu, S., Han, S. P., & Cho, D. [Ask for Reviews at the Right Time: Evidence from Two Field Experiments](#). *Journal of Marketing* (November 2022).

Universities: Lee School of Business, University of Nevada - Las Vegas; USC-SJTU Institute of Cultural and Creative Industry, Shanghai Jiao Tong University; W.P. Carey School of Business, Arizona State University and College of Business, Korea Advanced Institute of Science & Technology.

[Full Ariyh research summary](#)

PRO TIP: Reply to all reviews

Make sure you [reply to all reviews](#) you receive (positive or negative). An analysis of reviews on TripAdvisor and Expedia found that when companies started responding to reviews, their star ratings increased by an average of 0.12 (out of 5) and they received 12% more reviews.

PRO TIP: Frame your product as an experience

To increase reviews, focus on the [emotional or social experience around your product](#), when describing it (e.g. people socially drinking coffee) - rather than on its features (e.g. coffee bean origins). When framed this way, similar products on Amazon have more reviews.



Handling Complaints

How to respond to public complaints



Recommendation

- When you receive complaints on social media or review websites (e.g. Google, Tripadvisor), you are responding for everyone who reads it, not only the complainer.
- The best way to handle it, is to reply as soon as possible - but only once. Show empathy for the situation and try to give an explanation if possible. Then provide options to contact you privately and urge the complainer to do so.
- If you receive a large or regular flow of complaints, try setting up separate accounts for complaints and queries, to avoid crowding out your regular posts.



How it works

- Replying to complaints on social media reduces their negative effects, but can give more visibility to the complaint, affecting others' perception of your brand.
- Some social media sites, like Twitter and LinkedIn, give significantly more visibility to posts that you reply to, so others will see the complaint and your response.



Research

Scientific research by Golmohammadi, A., Havakhor, T., Gauri, D. K., & Comprix, J. [Complaint publicization in social media](#). *Journal of Marketing* (July 2021).

Universities: Belk College of Business, University of North Carolina - Charlotte; Fox School of Business, Temple University; Sam M. Walton College of Business, University of Arkansas; Martin J. Whitman School of Management, Syracuse University.

[Full Ariyh research summary](#)

PRO TIP: Use humor to reply to rude complaints

When you get a [rude public complaint on social media](#), respond using humor about the situation or complaint (not about the person complaining). Other people who see the complaint are more likely to engage - giving positive visibility to your answer - and become 18.3% more likely to buy from you compared to if there's no response. Make sure your humor is funny.

“ 5.

Referrals

Frame referrals as altruistic



Recommendation

- When asking customers to refer you to a friend, focus on an altruistic incentive for that friend (e.g. give \$20 off to a friend, give a free [offer] to a friend). You should still give an incentive to the referrer (e.g. you also get \$20 off your next purchase), but that should not be the focus of your message.
- In experiments, people were up to 60% more likely to make at least one referral if the referral incentive had an altruistic framing. Overall referrals increased 86%.



How it works

- People normally recommend a product to a friend because they think the friend would like it. It's an altruistic behavior.
- But when an incentive for the referrer is added (e.g. get \$20 if a friend joins) the incentive clashes with the altruistic motivation of recommending a product.
- Focusing the framing of the message on further helping the friend removes the guilt of receiving referral rewards.



Research

Scientific research by Jung, J., Bapna, R., Golden, J.M. and Sun, T. [Words matter! Towards a prosocial call-to-action for online referrals: Evidence from two field experiments.](#)
Information Systems Research (May 2020).

Universities: Fox School of Business and Management, Temple University; University of Minnesota; Collage.com and Marshall School of Business, University of Southern California.

[Full Ariyh research summary](#)

How to maximize referrals

- [Pre-fill referral messages](#) with optimal information when customers refer a friend. Information that the referrer already bought the product increases the likelihood of a conversion by 11%. Information about a referral reward for referring others, increased further referrals by 53%. Choose one or the other, don't include both.
- When possible, [avoid giving money](#) or cash discounts as rewards. Instead, try to offer a free gift, an upgrade, or other non-monetary promotion.
- [Very high referral rewards](#) boost your referral rate, but greatly reduce the quality of referrals - which makes your acquisition unprofitable. For example, the average profitability of referred customers for an online bank dropped 48% when the referral reward increased from €20 to €50.

NOTE

For more pricing and promotions insights - including which price digits to use for your price (just-below \$9.99 vs rounded \$10) and step-by-step instructions on how to price your product, please refer to [Ariyh's Pricing & Promotions playbook](#).

“ 6.

Prices, Discounts & Promotions

How To Display Prices

Show the price underneath your product

Recommendation

- Display your price below the product. Prices shown below (vs above the product) are perceived as lower, increasing people's likelihood of buying.
- Experiments found that people considered the same price (\$2.49) for dental floss to be 9% lower when it was positioned below the product. In a physical liquor store, sales were 35.2% higher when prices were shown below the bottles.

How it works

- The metaphor that “down = less” and “up = more” is strongly ingrained in most people. Seeing prices in a lower position triggers this association, making us consider the price lower than if displayed in a higher position.

Research

Scientific research by Barone, M., Coulter, K., & Li, X. [The upside of down: Presenting a price in a low or high location influences how consumers evaluate it](#). *Journal of Retailing* (September 2020).

Universities: University of Louisville; Clark University; Dell Technologies.

[Full Ariyh research summary](#)

PRO TIP: Show prices in red to men

If your audience is mostly male, [display your prices in red](#). Men will perceive them as a better deal. In experiments, men considered prices 66% lower when they were in red. Prices in red don't have any impact on women or when the price is closely analyzed (e.g. for important or expensive purchases).

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How to Frame Discounts

When to use amount (\$10) vs percentage (10%)



Recommendation

- Use a dollar amount discount (e.g. \$15 off) if the original price is higher than 100.
- Use a percentage discount (e.g. 20% off) if the price is less than 100 or less.
- For example, people were 16.3% more willing to buy a 480 pesos jacket showed a “120 pesos off” discount vs a “25% off” discount.



How it works

- Mathematically, the number used in the discount looks bigger. Bigger discount numbers instinctively seem better, and people don't always put in the mental effort to calculate it.
- For example, 20% off \$400 is \$80 off. 80, is a much bigger number than 20, so it's more attractive.

Product type	Discount		Buying Decision
	Percentage	Amount	
40	25%	\$10	Percentage (25%)
140	25%	\$35	Amount (35%)



Research

Scientific research by González, E. M., Esteva, E., Roggeveen, A. L. and Grewal, D. [Amount off versus percentage off - when does it matter?](#) *Journal of Business Research* (August 2015).

Universities: EGADE Business School and Babson College.

[Full Ariyh research summary](#)

PRO TIP: Give the optimal discount coupons

When giving discount coupons, [use a percentage discount \(e.g. 20% off\) if the coupon can be used for any item in your store](#). People will buy more. Use a dollar amount if the coupon is brand or product specific. People will be more likely to convert.

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Calculate your discount % using the discounted price



Recommendation

- Use the already discounted price, not the original price to calculate the discount amount.
- For example, if discounting by \$2 a full price of \$10, you would normally say 'now 20% lower' ($\$2/\$10 = 20\%$ lower). Instead, use the already discounted price of \$8 ($\$10 - \2 discount) and say 'was 25% higher' ($\$2/\$8 = 25\%$ higher). It's now a larger %.



How it works

- People pay more attention to the number in a discount than the surrounding text, and the larger the number is, the more likely people are to buy the item.



Research

Scientific research by Guha, A., Biswas, A., Grewal, D., Verma, S., Banerjee, S. and Nordfält, J. [Reframing the discount as a comparison against the sale price: does it make the discount more attractive?](#) *Journal of Marketing Research* (June 2018).

Universities: University of South Carolina; Wayne State University; Babson College; Lawrence Technological University and Stockholm School of Economics.

[Full Ariyh research summary](#)



How To Frame Promotions

Describe promotions effectively

- When framing your promotions, [describe the restriction first, then the offer](#) (e.g. “Spend \$100 and get \$20 off” instead of “Get \$20 off by spending \$100”). Doing this will make the promotion seem like a reward, rather than a restriction.
- Frame a promotion as [a free gift rather than a bundle deal](#) to cut product returns by up to 50%. Market a promotion as “Buy X and get Y free” rather than “Buy X and Y for a special price”.
- Make the main [product in a bundle promotion a gift](#) to maximize sales. For example, for customers searching for shampoo, offer a promotion to “Buy a conditioner and get a shampoo free” rather than “Buy a shampoo and get a conditioner free”. They will feel lucky and be more likely to buy. Sales were up to 78% higher in experiments.

Invent 'Special Days' for your Promotions



Recommendation

- Promotions are more effective if they are tied to a special day. The special day must be creative (e.g. a promotion on World Cotton Day is creative; Father's Day is not). Non-creative or inappropriate day-themed promotions don't help.
- For example, people were 25% more likely to buy from a picnic-related retailer when the one-day promotion was framed as "National Picnic Day Sale" (vs "Annual One Day Sale").



How it works

- People notice the creativity of special day promotions that are original and appropriate.
- The unexpected unique creativity makes them feel better toward the brand and increases enjoyment, making them more likely to buy.

Some options for creative special promotion days

Cross-Industry	Fashion / Apparel	Electronics / Tech	Food
Company founding day Customer anniversary Customer birthday	March 6: National Dress Day July 5: International Bikini Day August 9: International Laundry & Textile Service Day September 24 - October 2: Wool Week October 7: World Cotton Day October 13: International Suit Up Day	February 13: World Radio Day March 9: National Dishwasher Appreciation Day April 25: National Telephone Day May 17: National Baking Day November 21: World Television Day	February 9: Meat Day April 4: International Carrot Day May 21: International Tea Day June 1: International Milk Day June 15: World Meat Free Day July 7: World Chocolate Day October 1: World Vegetarian Day & International Coffee Day October 25: World Pasta Day November 1: World Vegan Day



Research

Scientific research by Zane, D. M., Reczek, R. W. and Haws, K. L. [Promoting Pi Day: Consumer responses to special day-themed promotions](#). *Journal of Consumer Psychology* (September 2021).

Universities: College of Business, Lehigh University; The College of Business, Ohio State University and Owen Graduate School of Management, Vanderbilt University.

[Full Ariyh research summary](#)

Time-limited promotions aren't as strong as we thought



Recommendation

- Limited time promotions (e.g. deal expires in 3 days) are less effective at driving sales online than they are offline (e.g. in stores). They drive awareness and engagement (e.g. email open, website visit), but are often weak at driving actual immediate sales.
- To increase their strength, justify the time pressure (e.g. customers' birthday, seasonal sale) and use a short time window (e.g. 24 hours rather than one week).



How it works

- People see limited offline availability - scarcity - as a signal that something is valuable.
- Online however, they perceive it as an artificial sense of urgency used to persuade them, so they may feel manipulated and distracted from the value of the deal.



Research

Scientific research by Hmurovic, J., Lamberton, C. and Goldsmith, K. [Examining the efficacy of time scarcity marketing promotions in online retail](#). *Journal of Marketing Research* (July 2022).

Universities: LeBow College of Business, Drexel University; The Wharton School, University of Pennsylvania, and Owen Graduate School of Management, Vanderbilt University.

[Full Ariyh research summary](#)

Charge a nominal amount for optional upgrades



Recommendation

- Promotions that offer an upgrade (e.g. a larger size or add-on item) are more attractive when the upgrade is charged a very small token amount (e.g. 1¢), rather than for free.
- In experiments, people rated a deal that gave a digital camera an upgraded memory card 39% better when the upgrade cost ¥0.1 (vs free). Similarly, 17.1% more people decided to buy a larger size can of juice when the upgrade cost ¥0.1 (vs free).



How it works

- Normally, people judge prices and deal using reference points and comparisons. When something is free, people don't use \$0 as a reference point - they make their judgment based on the price of the main product.
- If a very small token cost is attached to an upgrade, people look at that cost, compare it to the benefit they receive, and judge it a better deal.



Research

Scientific research by Mao, W. [Sometimes “fee” is better than “free”: Token promotional pricing and consumer reactions to price promotion offering product upgrades](#). *Journal of Retailing* (May 2016).

Universities: School of Business, Southwestern University of Finance and Economics (China).

[Full Ariyh research summary](#)



Thank you

Thank you for reading and purchasing Ariyh's Science-based Ecommerce Optimization Playbook.

Did you find the playbook useful? I'd be very grateful if you **leave a review**, so others know what to expect. Have suggestions for improvements? **I would love to hear from you**. Email me at thomas@ariyh.com.

Want to **optimize specific areas of your business**? Check out [Ariyh's other Playbooks](#) (new ones will be released in the future).

Interested in my **hands-on advice to help you optimize your Ecommerce store**? You can [reach out here](#). Here's what some previous businesses that I helped have to say:



Our conversion rate increased by 15% after implementing the recommendations. Thomas's demeanor, process, and final product were world-class.



William Gallmeyer - President
Green Projects Group



I highly recommend Ariyh and Thomas. One of his first recommendations we tested already increased conversions on our homepage by 6.3%.



Joel Azzopardi - CEO
Freddystore.com



We saw an almost 20% jump in customer profitability. Thomas quickly understood our business, found opportunities, and gave us excellent recommendations.



Tabish Raza - Co-Founder
Increw